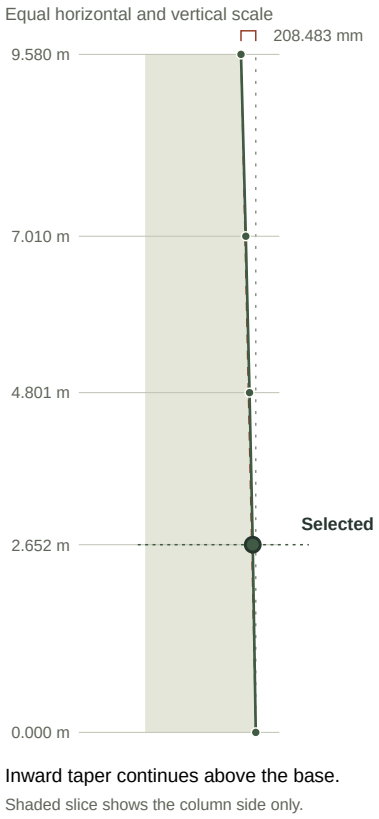
The curve inside the taper

Penrose’s five corrected offsets from one exterior flute edge of the fourth east-front Parthenon column from the northeast corner. Source: *An Investigation of the Principles of Athenian Architecture*, second edition (1888), p. 41.

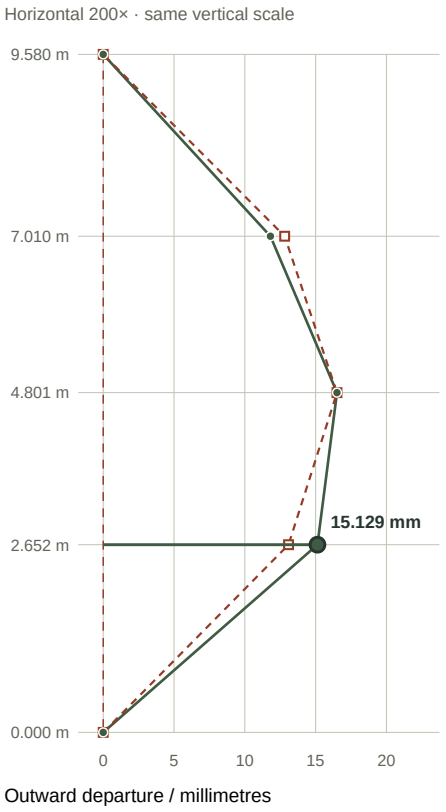
Selected height: **2.652 m**. Horizontal departure enlargement: **200×**. Prediction: **Penrose’s circle**. Building comparison: **equal shaft length**.

Measured offsets, equal scales



Axis-referenced offsets remove the whole column’s inward lean. The shaded slice does not reconstruct its full diameter.

The straight taper subtracted



Horizontal distances 200×; the vertical scale matches the first drawing. Zero is the straight endpoint taper.

Green circles are corrected column offsets; orange dashes show the straight taper. Orange squares are Penrose’s printed circle predictions. Straight guides connect the five stations; they are not intermediate measurements or a reconstruction of the continuous mathematical curve.

At the selected station

57.709 mm straight taper − 42.581 mm corrected column = **15.129 mm outward departure**.

The chord offset is 0.684 × 8.70 / 31.43 feet. All higher measured stations lie inward of the base. Entasis is an addition relative to straight taper; it need not make the shaft wider than its base.

Height / ft	Column / ft	Straight taper / ft	Departure / ft	Departure / mm	Circle / ft	Prediction − column / mm
0.00	0.0000	0.000000	0.000000	0.000	0.0000	0.000
8.70	0.1397	0.189335	0.049635	15.129	0.1464	+2.042
15.75	0.2886	0.342762	0.054162	16.508	0.2886	0.000
23.00	0.4618	0.500541	0.038741	11.808	0.4585	-1.006
31.43	0.6840	0.684000	0.000000	0.000	0.6840	0.000

Original offsets are in English feet. Metric conversion uses 304.8 mm per foot. Additional calculated decimals are not estimates of measurement accuracy; the source gives no statistical uncertainty for these rows.

What the predictions establish

The circle passes through the base, middle and top stations by construction. Its largest discrepancy in these rows is 0.0067 feet (2.042 mm), at 8.70 feet. The hyperbola's largest printed discrepancy is 0.0011 feet (0.335 mm), at 15.75 feet. Penrose's prose says no more than 0.001 feet, but his printed middle row is 0.2875 against 0.2886. This document retains the numbers.

Penrose inferred an ancient hyperbolic design from the agreement. A close fit alone does not establish the builders' method. His optical-correction explanation is also disputed; see Alain Goriely's 2026 analysis. No optical experiment is performed here.

Six reported maxima

The largest departure among our five stations is 16.508 mm at 15.75 feet. Penrose separately reports 0.057 feet (17.374 mm) at 13.8 feet. A maximum in a short table is not necessarily the maximum of the full profile.

Selected comparison: each shaft scaled to the Parthenon's 31.43-foot length. Normalization is maximum × 31.43 / shaft length. Penrose's separately rounded comparison remains beside the calculated result.

Name	Shaft / ft	Maximum / ft	Height above stylobate / ft	Equal-length maximum / ft	Printed comparison / ft	Selected display
Erechtheum · north portico	21.12	0.0195	10.42	0.029019	0.0290	8.845 mm
Theseum	17.10	0.0230	8.70	0.042274	0.0420	12.885 mm
Parthenon · peristyle	31.43	0.0570	13.80	0.057000	0.0570	17.374 mm
Propylaea · small order	17.50	0.0343	9.33	0.061603	0.0615	18.777 mm
Propylaea · large eastern order	25.60	0.0627	13.75	0.076979	0.0770	23.463 mm
Jupiter Olympius	43.70	0.1180	18.40	0.084868	0.0840	25.868 mm

The shaft lengths exclude the apophyge and related details; heights of maxima are measured above the stylobate. These are reported maxima, not six reconstructed continuous profiles. Historical building names are retained.

An addition in the middle parts

de adiectione, quae adicitur in mediis columnis, quae apud Graecos εντασις appellatur, in extremo libro erit forma et ratio eius, quemadmodum mollis et conveniens efficiatur, subscripta.

Concerning the addition made in the middle parts of columns, which the Greeks call entasis, a diagram and an explanation of how to make it gentle and appropriate will be set down at the end of the book.

Vitruvius III.3.13, Krohn (1912), p. 68. New English translation. Greek εντασις follows this printing's unaccented word; the dictionary headword below supplies accents. Line-end hyphens are joined. The promised ancient illustration has not survived; neither this sentence nor our drawing recovers its recipe.

ἔντασις

entasis · tension, stretching

A feminine noun, genitive ἐντάσεως. In the architectural passage it names the addition within a tapering shaft. Its broader meaning does not establish the architect's motive. Dictionary.

ἐντείνω

enteinō · I stretch, strain tight

The related verb helps remember the noun. Penrose connects the word with stretching a line or bending a bow; that is an explanation of the name, not a measured optical effect. Dictionary.

de adiectione

concerning the addition

De takes the ablative; adiectione is the ablative singular of adiectio. The related passive verb adicitur means ‘is added’.

forma et ratio

a diagram and an explanation

Here forma is the promised figure and ratio its account or method. Reading ratio as a supplied numerical proportion would invent information absent from this sentence.

in mediis columnis

in the middle parts of columns

Mediis and columnis are ablative plural. The phrase gives a region; it does not specify a maximum at exactly half the shaft's height.

mollis et conveniens

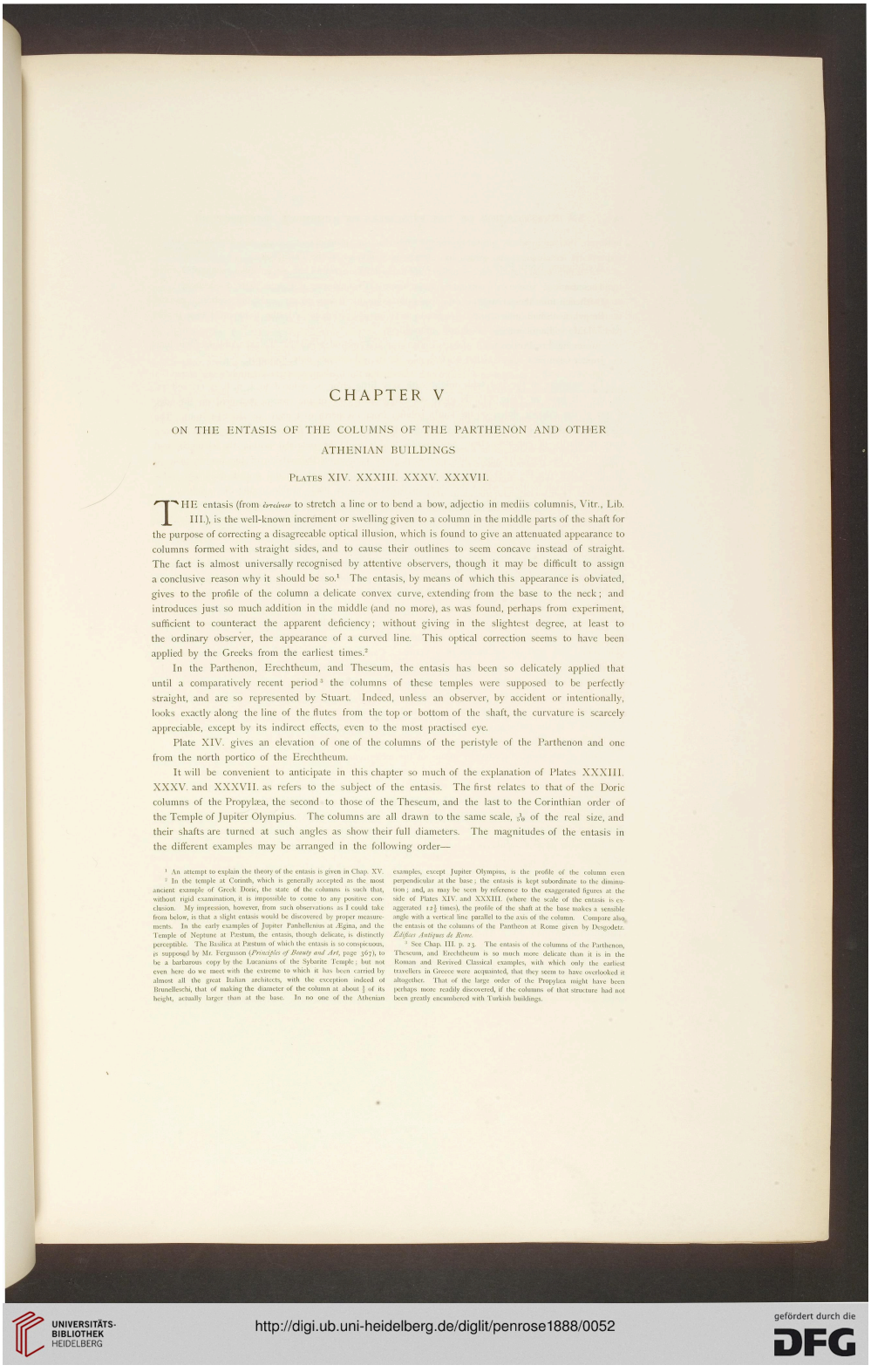
gentle and appropriate

The adjectives describe the intended result. Quemadmodum introduces how it is to be made; efficiatur is a passive subjunctive.

The printed evidence

Four Penrose pages from Heidelberg University Library and one Krohn page from Internet Archive. All five public-domain images are embedded below. Penrose's p. 40 explains the wire measurements, drum-shift corrections and change of reference line; the plotted numbers come from his corrected five-row comparison on p. 41.

Penrose · p. 39. The meaning of entasis, including Penrose's observation that these profiles remain narrower above the base. Library record.



CHAPTER V

ON THE ENTASIS OF THE COLUMNS OF THE PARTHENON AND OTHER ATHENIAN BUILDINGS

PLATES XIV. XXXIII. XXXV. XXXVII.

THE entasis (from ἐντείνειν to stretch a line or to bend a bow, adjectio in mediis columnis, Vitr., Lib. III.), is the well-known increment or swelling given to a column in the middle parts of the shaft for the purpose of correcting a disagreeable optical illusion, which is found to give an attenuated appearance to columns formed with straight sides, and to cause their outlines to seem concave instead of straight. The fact is almost universally recognised by attentive observers, though it may be difficult to assign a conclusive reason why it should be so.[1] The entasis, by means of which this appearance is obviated, gives to the profile of the column a delicate convex curve, extending from the base to the neck; and introduces just so much addition in the middle (and no more), as was found, perhaps from experiment, sufficient to counteract the apparent deficiency; without giving in the slightest degree, at least to the ordinary observer, the appearance of a curved line. This optical correction seems to have been applied by the Greeks from the earliest times.[2]

In the Parthenon, Erechtheum, and Theseum, the entasis has been so delicately applied that until a comparatively recent period[3] the columns of these temples were supposed to be perfectly straight, and are so represented by Stuart. Indeed, unless an observer, by accident or intentionally, looks exactly along the line of the flutes from the top or bottom of the shaft, the curvature is scarcely appreciable, except by its indirect effects, even to the most practised eye.

Plate XIV. gives an elevation of one of the columns of the peristyle of the Parthenon and one from the north portico of the Erechtheum.

It will be convenient to anticipate in this chapter so much of the explanation of Plates XXXIII. XXXV. and XXXVII. as refers to the subject of the entasis. The first relates to that of the Doric columns of the Propylæa, the second to those of the Theseum, and the last to the Corinthian order of the Temple of Jupiter Olympius. The columns are all drawn to the same scale, $\frac{3}{16}$ of the real size, and their shafts are turned at such angles as show their full diameters. The magnitudes of the entasis in the different examples may be arranged in the following order—

[1] An attempt to explain the theory of the entasis is given in Chap. XV.

[2] In the temple at Corinth, which is generally accepted as the most ancient example of Greek Doric, the state of the columns is such that, without rigid examination, it is impossible to come to any positive conclusion. My impression, however, from such observations as I could take from below, is that a slight entasis would be discovered by proper measurements. In the early examples of Jupiter Panhellenius at Ægina, and the Temple of Neptune at Pæstum, the entasis, though delicate, is distinctly perceptible. The Basilica at Pæstum of which the entasis is so conspicuous, is supposed by Mr. Fergusson (*Principles of Beauty and Art*, page 367), to be a barbarous copy by the Lucanians of the Sybarite Temple; but not even here do we meet with the extreme to which it has been carried by almost all the great Italian architects, with the exception indeed of Brunelleschi, that of making the diameter of the column at about $\frac{1}{3}$ of its height, actually larger than at the base. In no one of the Athenian examples, except Jupiter Olympius, is the profile of the column even perpendicular at the base; the entasis is kept subordinate to the diminution; and, as may be seen by reference to the exaggerated figures at the side of Plates XIV. and XXXIII. (where the scale of the entasis is exaggerated 12½ times), the profile of the shaft at the base makes a sensible angle with a vertical line parallel to the axis of the column. Compare also the entasis of the columns of the Pantheon at Rome given by Desgodetz. *Edifices Antiques de Rome.*

[3] See Chap. III. p. 23. The entasis of the columns of the Parthenon, Theseum, and Erechtheum is so much more delicate than it is in the Roman and Revived Classical examples, with which only the earliest travellers in Greece were acquainted, that they seem to have overlooked it altogether. That of the large order of the Propylæa might have been perhaps more readily discovered, if the columns of that structure had not been greatly encumbered with Turkish buildings.

UNIVERSITÄTS-BIBLIOTHEK HEIDELBERG
http://digi.ub.uni-heidelberg.de/diglit/penrose1888/0052
gefördert durch die DFG

Buildings.	Length of shaft exclusive of Apophyge, etc.	Maximum entasis.	Height of the maximum entasis above the stylobate.	Comparative entasis in a shaft 31.43 feet in length.
Erechtheum, North Portico	21.12	·0195	$10.42 = \frac{5}{12}$ of the height of column.	.029
Theseum	17.1	.023	$8.7 = \frac{1}{2}$ shaft	.042
Parthenon from the peristyles	31.43	.057	$13.8 = \frac{3}{8}$ column	.057
Propylæa—Small order	17.5	.0343	$9.33 = \frac{1}{2}$ column	.0615
Ditto Large order from Eastern Portico	25.6	.0627 or 1.0031 = .0633	$13.75 = \frac{1}{2}$ column	.077
Jupiter Olympius	43.70	.118	$18.4 = \frac{1}{3}$ height of the column	.084

The numbers in the last table might with very little error be reduced to 4, 6, 8, 9, 11, and 12. On the left hand of Plate XIV. is given an exaggerated view of the entasis, measured on the edge of one of the exterior flutes, of the fourth column from the north-east angle, on the east front. The edges of the flutes in those columns which had been best preserved from injury, were found to have more regularity of curvature than any profiles which could be obtained within the channels, and moreover to be themselves generally somewhat more regular on the outside of the temple than on the inside towards the cella walls, which parts being usually in shade, did not demand the same absolute precision as those where the slightest defect would be made visible by the cast shadow.

I have given in the plate the best preserved specimen I could find; but the exactness of the curvature in this case is by no means peculiar, as may be seen by the other examples given in the Appendix, Art. 16.

The entasis, as shown in Plates XIV. and XXXIII. and XXXVII., was measured in the following manner. The edge or arris of a flute at the bottom of the shaft was first selected, or its position recovered, if defective, by completing on the pavement the contour of the contiguous flutes, according to their ascertained law (see Plate XXI. Fig. 14). The edge was usually perfect at or near the neck. A fine harp wire was then brought as near as was convenient to these two points, and strained by means of weights as tight as it would bear. The offsets between the wire and the most perfect surfaces presented by the column were then measured with a slide joint, and read off by means of a vernier. Care was taken to obtain, whenever possible, two measurements in each stone, and the amount of the shifts at the joints was also ascertained. By this means, since generally no movements, excepting horizontal shifts, can have taken place (see Chap. III. Sect. 2), if the surface be sufficiently well preserved, of which, in carefully selected places, there is little doubt, the exact form can be recovered.

Table 1 in the plates gives the heights above the pavement where these offsets were taken.

Table 2 the offsets exactly as they were measured.

Table 3 the same, corrected by means of the measurements of the horizontal shifts.

In Table 4 the offsets which hitherto were measured horizontally from the strained wire, are transferred by calculation, so as to be measured (also horizontally) from an imaginary vertical straight line Y Y′ parallel to the axis of the column, and passing through the point Y′, which in the example before us from the Parthenon is the extremity of the lower diameter of the column. The amount of the inclination of the axis of the column is indicated by the plumb-lines in each case.

From the measurements here recorded we may endeavour to determine the nature of the curve employed.

I have found the entasis in every case so nearly resembling one of the forms of the conic sections, viz. the hyperbola (Fig. 2 Plate XLVII., see also Appendix, Art. 4), that I cannot doubt that this was the curve used in the columns of the Athenian structures. This is the more likely, because we may be satisfied that these curves, although their invention has been assigned by mathematical writers to a later age, were nevertheless certainly known, though perhaps not as sections of the cone, to the builders of the Parthenon, as will be shown when we come to the subject of the profiles of the mouldings in Chap. VII.;[1] and although perhaps some of these

[1] It is not unlikely that the curves of the second order of which we are speaking were known to the Greeks even at a much earlier age. The measurements given of the echinus of the capitals of the temple at Corinth (see Stuart and the *Expédition Scientifique de Morée*), and those of

UNIVERSITÄTS-BIBLIOTHEK HEIDELBERG
http://digi.ub.uni-heidelberg.de/diglit/penrose1888/0053
gefördert durch die DFG

examples of entasis from the extreme delicacy of their curvature might be successfully compared with other regular curves known to the ancients, such for instance as the conchoid of Nicomedes, Fig. 1, Plate XLVII., the entasis of the columns of the Propylæa, as far as I am aware, can be reduced to no known simple curve, except the hyperbola; and there is strong internal evidence that the columns of the Parthenon are of the same nature. In the Appendix, in the description to Plate XLVII., will be found a short description of the different curves employed, or supposed to be employed in the Athenian buildings; and I must beg the reader to refer thither for their definitions, and also for the names of their different parts, etc., which I shall use here without further explanation.

To begin with the entasis of the Parthenon.

In searching out the nature of this curve, it may be worth while to try how nearly it coincides with a circular arc passing through the upper, lower, and middle points.[1] It may be shown that the curvature of the column is much quicker than that of the circle in the lower, and flatter in the upper half, and the circle gives no satisfactory approximation; neither would an ellipse nor even a parabola, applied as usual, at their vertices materially alter the case. Had the corresponding segment of a parabola been used instead of the circle, the approximation would have been carried only .0003 nearer; and even the conchoid of Nicomedes, unless placed obliquely, would fail to give a satisfactory resemblance; but if the hyperbola be tried, we find not only a wonderful agreement at all points, but the geometrical axis of the curve is parallel to the axis of the column. In Table 5 in the plate, in which are given the ordinates to an hyperbola obtained by calculation, we find the following correspondence of measurements—

Height.	Offsets of Hyperbola.	Offsets to Column.
0	0	0
8.7	.1392	.1397
15.75	.2875	.2886
23.0	.4613	.4618
31.43	.684	.684

none of which differ more than one thousandth part of a foot, which is just the space occupied from centre to centre in Plate XIV. by the lines drawn to indicate the fillet between the flutes of the Ionic example. In the other measurements the correspondence is scarcely less complete, and we may be satisfied that we have here as perfect identity as it is practicable to attain.[2]

The entasis of the columns of the Posticum (Inner order) seems to be identical with those of the peristyles. An example of one of these columns will be given in the Appendix, Art. 17.

In the column of the northern portico of the Erechtheum on the right hand of the plate, which also is drawn so as to present its full diameter, the entasis has been measured precisely in the same manner as in the previous example. The offsets are transferred to the straight line YY',[3] and are compared

the Temple of Jupiter Panhellenius, Ægina, and of Jupiter at Olympia by the last mentioned authority, also the sections from the early examples at Athens given in page 6 of this work, point to one of the conic sections as the form probably used in those early buildings.

[1] We may find by calculation what is not manageable in practice in geometry, the radius of a circle passing through the above-mentioned points, namely coinciding with the column at the heights 0. 15.75. and 31.43 (see Table 1 in the plate). The radius of such a circle would be 2280 feet; which dimension, just 10 times the length of the flank of the temple, may indeed point to some analogy followed in the selection of the amount of the entasis; but the offsets obtained from this circle differ very considerably from its curvature; for if we compare them at the nearly intermediate heights, 8.7 and 23.0, the result which we obtain is as follows—

Height.	Offsets in the circle.	Offsets from the entasis of the column.
0	0	0
8.7	.1464	.1397
15.75	.2886	.2886
23.0	.4585	.4618
31.43	.684	.684

[2] Besides the correspondence of the measurements, the relations among the elements of this curve are such as to give great additional confirmation.

The rectangular axes of this figure are vertical and horizontal, that is, parallel to the architectural lines of the column.

The principal lines is 1 Attic foot = 1.0134 foot.

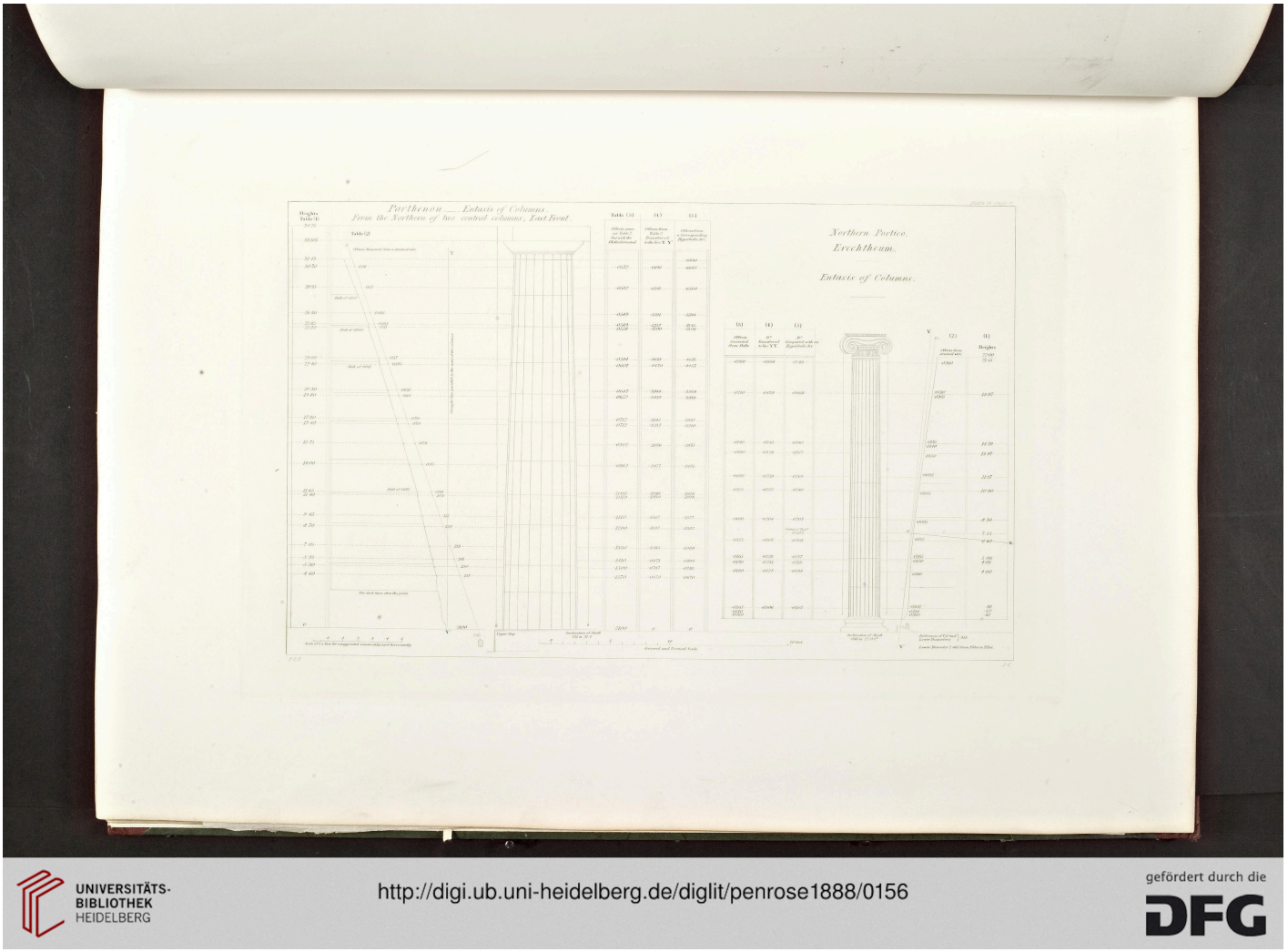
The eccentricity, or constant ratio, is as the integral number 30 to 1, it may be stated in addition, that the radius of curvature of this hyperbola at the point coinciding with the base of the column is five times the length of the building. The distance of the vertex below the base is just 2 diameters, and the maximum entasis, which is exactly $\frac{1}{30}$ of the half diminution of the column, is produced by this hyperbola at the height of 13.8 feet above the stylobate, just $\frac{2}{3}$ of the height of the column.

The value of the proofs of design in the relations among the elements of the curve arises from the fact that in consequence of the absence, in the segment employed in the case of the Parthenon, of the vertex of the curve (or the point on each side of which the curvature changes symmetrically), and its extreme delicacy, it might be possible to find other regular curves which should coincide almost as well with the measurement as that from which Table 5 is calculated. But there is no such ambiguity respecting the nature of the curvature used in the columns of the Propylæa and Erechtheum, for in these examples the vertex occurs near the middle of the shaft, and the curves can be proved to be hyperbolas, having also generally some simple relations among their elements.

[3] This straight line is not, as in the former instance, perpendicular, but lies obliquely with regard to the axis of the column. YY' is in fact the

UNIVERSITÄTS-BIBLIOTHEK HEIDELBERG
http://digi.ub.uni-heidelberg.de/diglit/penrose1888/0054
gefördert durch die DFG

Penrose · plate XIV. The fuller survey of the northern of the two central east-front columns, and a separate Erechtheum column. The original plate already enlarges the horizontal entasis. Only the five-row p. 41 comparison is digitized here. Library record.



UNIVERSITÄTS-
BIBLIOTHEK
HEIDELBERG
http://digi.ub.uni-heidelberg.de/diglit/penrose1888/0156
gefördert durch die
DFG

Vitruvius · p. 68. Krohn's 1912 Latin edition, III.3.13. The sentence naming entasis appears immediately above chapter IV; the critical apparatus remains visible below. Library record.

autem propter altitudinis intervallum scandentis oculi species adiciuntur crassitudinibus temperaturae. venustates enim persequitur visus, cuius si non blandimur voluptati proportione et modulorum adiectionibus, uti quod fallitur temperatione adaugeatur, vastus et invenustus conspicientibus remittetur aspectus. de adiectione, quae adicitur in mediis columnis, quae apud Graecos ενταcιc appellatur, in extremo libro erit forma et ratio eius, quemadmodum mollis et conveniens efficiatur, subscripta.

IV Fundationes eorum operum fodiantur, si queat inveniri, ad solidum et in solido, quantum ex amplitudine operis pro ratione videbitur, extruaturque structura totum solum quam solidissima. supraque terram parietes extruantur sub columnas dimidio crassiores quam columnae sunt futurae, uti firmiora sint inferiora superioribus; quae stereobates appellantur, nam excipiunt onera. spirarumque proiecturae non procedant extra solidum; item supra parietis ad eundem modum crassitudo servanda est. intervalla autem concama-
2 randa aut solidanda festucationibus, uti distineantur. sin autem solidum non invenietur, sed locus erit congesticius ad imum aut paluster, tunc is locus fodiatur exinaniaturque et palis alneis aut oleagineis ⟨aut⟩ robusteis ustilatis configatur, sublicaque machinis adigatur quam creberrime, carbonibusque expleantur intervalla palorum, et tunc structuris solidissimis fundamenta impleantur. extructis autem fundamentis ad libramentum stylobatae sunt conlocandi.
3 supra stylobatas columnae disponendae, quemadmodum supra scriptum est, sive in pycnostylo, quemadmodum

---

1 oculi *rec*: oculis *x* species *x* (*genit. ut* materies *60, 12 et* notities *133, 9*) 7 entasis *HG*, enthasis *S* 8 forma et *Wesseling* (*cf. 80, 5*): formata *x* 11 ad solidum et in solido *Phil* (*cf. 21, 3*): ab solido et in solidum *x* 15 est (ē) ereobates *x* 17 solium *H* parietis *Joc*: -tes *x* 18 concameranda *S* 19 distineantur *Joc*: destineantur *x* 20 congesticius *G*: coniesticius *HS* 21 paluster *G*: plauster *S*, plai | ter *cum ras. H* 22 alneis *Joc* (*cf. 130, 22*): saligneis *x* *add. Joc* 23 configatur *G*: configuratur *HS* subligaque *x* 26 collocandi *Joc*: conlocandae (-ę) *x*

Open this view in Gonia. This file opens without a network connection. Code, new drawings and new English: MIT, Anton Soloviev, 2026. Penrose and Vitruvius text and scans: public domain. Library attribution is retained on the Penrose scans.